

COMPOUND PP

Heat/weathering resistant, non-flammable
Composite PP Resin

- ▶ FH42P
- ▶ FH51
- ▶ FH51G

● **Description**

Non-flammable PP, FH51 and FH51G, have a feature of high heat-resistance and rigidity, while FH42P can be used for out-door, for example X-Mas socket, needing heat/weathering resistance.

To be applied for Out-Door, the required standard is UL of non-flammability and eco-friendly feature. FH42P Grade keeps non-flammability of UL94 V-0 and fl grade of UL746C, while FH51 exhibits excellent non-flammability of UL94 V-2.

● **Characteristics**

- ▶ FH51: As non-flammability of UL94 V-2 grade, it is easy to form with excellent fluidity and widely used for electric/electronic parts including X-mas sockets.
- ▶ FH51G : As non-flammability of UL94 V-2 grade, it is easy to form with excellent fluidity and widely used for, in particular, extension sockets with its high heat resistance and rigidity.
- ▶ FH42P: As non-flammability of UL94 V-0 grade and eco-friendly feature of UL746C, it is easy to form with excellent fluidity and widely used for out-door with its distinctive heat/light resistance, hydrothermal resistance, and long-term thermal stability.

● **Applications**

- ▶ X-Mas socket(FH51, FH42P)
- ▶ Electric parts for In-Door and Out-Door needing non-flammability and weathering resistance (FH51, FH42P)
- ▶ Extension socket needing high heat-resistance and rigidity (FH51G)

● **Major Properties Requirement**

- ▶ UL94 V-0, V-2 Grade Non-flammability
- ▶ UL746C f1 Grade Eco-friendly feature
- ▶ High heat-resistance and rigidity
- ▶ IEC695-2-1 (850℃)

● **General Processing Guide**

- ▶ Heat/weathering resistant, non-flammable PP, FH51, FH51G, and FH42P, have a similar processing condition as general non-flammable PP. While separate drying before molding is not necessary, drying for about 2 hours at 90~100℃ helps to get better appearance of a molding product in processing.
- ▶ Use in high temperature causes dismantlement so that it is molded under 220℃ of the resin temperature.
- ▶ There would be no problem when molding in normal cycle time. In order to avoid dismantlement of non-flammable agents among residual resin, the residual resin in the cylinder should be purged and cleaned with flammable PP at both break and finish.

Standard PP processing conditions may be applied, and the typical processing conditions are as follows:

Conditions		Data
Cylinder Temp.	Feeding zone	170 ~ 180
	Plasticizing zone	180 ~ 200
	Metering zone	180 ~ 200
Nozzle Temp. (°C)		190 ~ 210
Mold Temp. (°C)		40 ~ 70
Injection Pressure(kg/cm ²)		400 ~ 800
Back Pressure (kg/cm ²)		5 ~ 20
Injection Speed (%)		50 ~ 80

● Physical Properties

► Resin Properties

Properties	Test method	Condition	Unit	FH51	FH51G	FH42P
Melt index	ASTM D1238	230℃	g/10min	12.0	8.0	10.0
Gravity	ASTM D792	-	-	0.93	0.97	0.98
Tensile strength at Yield	ASTM D638	50mm/min	kg/cm ²	350	335	340
Elongation at Break			%	30	40	120
Flexural Modulus	ASTM D790	50mm/min	kg/cm ²	17,000	21,500	18,000
IZOD Impact Strength	ASTM D256	23℃	kg.cm/cm	4.0	3.0	3.5
Heat Distortion Temperature	ASTM D648	4.6kgf	℃	120	128	118
Surface Hardness	ASTM D785	Rockwell	R-Scale	95	104	100
Mold shrinkage	Hanwha Total	2mm(t)	%	1.4~1.7	1.2~1.5	1.4~1.7
Non- flammability	UL94	0.75mm	-	V-2 (1/32")	V-2 (1/32")	V-0 (1/32")
	UL746B	0.75mm	-	120℃	120℃	120℃
	UL746C	0.75mm				
	IEC695-2-1	1.6mm	℃	850	850	960

* Note: The above data are for reference materials that injection molded test pieces are used to measure their physical property values, and are

● **Food Contact Application**

- ▶ Hanwha Total FH42P, FH51, FH51G are not intended for use in food-contact and medical applications either.
- ▶ In case you might need additional technical or regulatory information, please contact Hanwha Total Composite Development Team.

● **Other Information**

The information in this document can be used for reference only, not to be construed as specification. Customers are responsible for determine whether our product and information is suitable for their particular purpose and for the compliance with related law.

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